MIDWEST GEOLOGICAL
SEQUESTRATION CONSORTIUM

MVA

Monitoring, Verification, & Accounting

Near-Surface Monitoring of Stored Carbon Dioxide

The extensive Monitoring, Verification, and Accounting (MVA) program undertaken for the Illinois Basin - Decatur Project (IBDP) is being conducted in three phases, pre-, concurrent, and post-injection. The research-based MVA program will inform commercial projects underway such as the Illinois Industrial Carbon Capture and Storage (IL-ICCS) Project and new projects coming online in the US and globally.

Near-surface and deep subsurface monitoring techniques are integral to meeting the MVA and project goals of:

1) Establishing pre-injection conditions to evaluate potential impacts from carbon dioxide (CO_2) injection,

2) Demonstrating that project activities are protective of human health and the environment, and

3) Quantifying pressure changes and tracking stored CO_2 in the Mount Simon reservoir.







Near-surface monitoring efforts include near-infrared color aerial imagery acquisition, surface deformation monitoring, net CO_2 flux monitoring, soil CO_2 flux monitoring, soil gas sampling, high-resolution electrical earth resistivity surveys, atmospheric CO_2 flux, meteorological monitoring, and shallow groundwater sampling.

Deep subsurface monitoring efforts include 2D seismic surveying; 3D seismic surveying and vertical seismic profiling; passive seismic monitoring; injection zone temperature, pressure, and fluid monitoring; above caprock temperature, pressure, and fluid monitoring; and open and cased hole logging. For more information see the deep monitoring brochure.

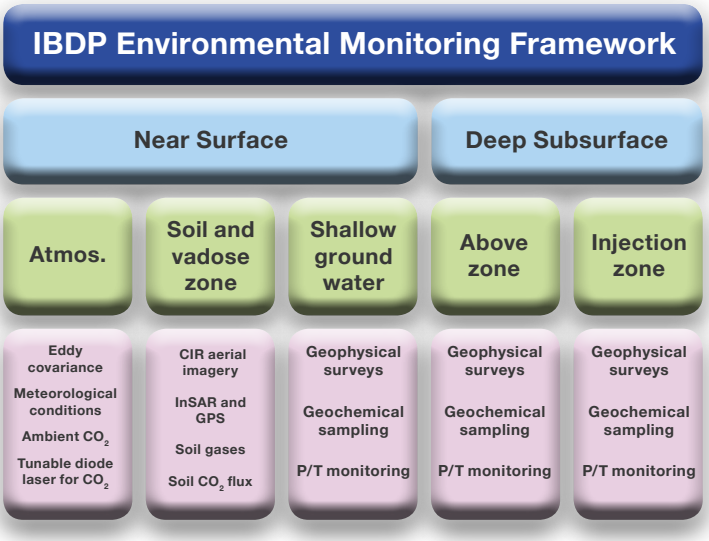
Atmospheric Monitoring

An eddy covariance (EC) tower has been deployed and is monitoring net CO_2 flux in the atmosphere along with other parameters that can affect ambient CO_2 movement. The EC tower collects continuous data from an automated system and 6 months of pre-injection baseline data have been obtained. These data are important to understand the dynamics of CO_2 movement at the IBDP site.

Soil Monitoring

In natural ecological systems, soil and vadose zone CO_2 concentrations and fluxes are controlled by multiple factors, such as (i) root respiration, (ii) organic carbon biodegradation in soil, (iii) groundwater CO_2 outgassing, (iv) CO_2 uptake by photosynthesis, and (v) exchange of CO_2 with the atmosphere. Detection of any CO_2 leakage into the near-surface environment is critical to successful validation of CO_2 storage and can be challenging due to large natural variability of soil CO_2 concentrations over space and time.

Soil flux and soil gas sampling data from the IBDP site, collected over a 24-month pre-injection period, provides a baseline from which any significant environmental changes may be detected during the injection and post-injection periods. Analysis of soil gas compositions and isotopic signatures of pre-injection and injection period data show that no anomalies related to CO_2 injection have been observed.



Illinois Basin - Decatur Project Environmental Monitoring Framework.

Shallow Groundwater Monitoring

Beginning in October 2008, seventeen shallow groundwater monitoring wells ranging in depth from 20 to 300 feet (6 to 91 meters) have been installed and instrumented to monitor water levels and chemistry.



Eddy covariance monitoring system.





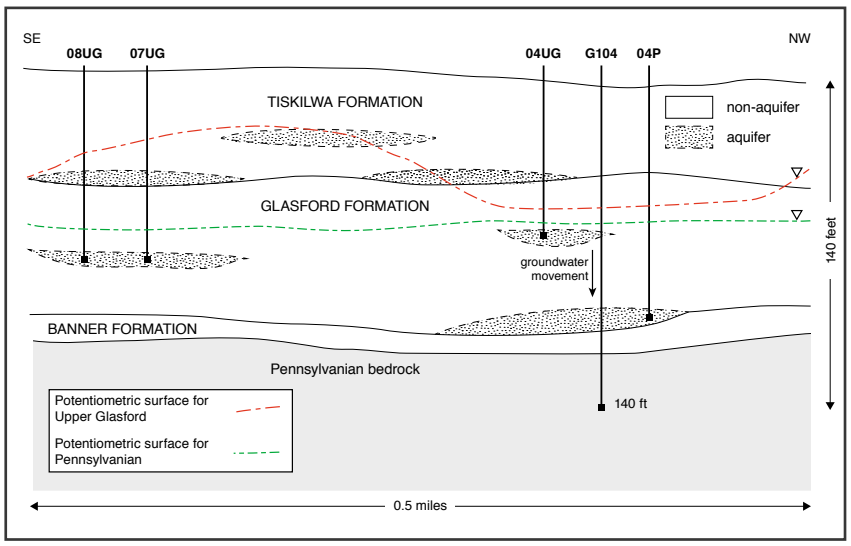
Collecting a soil flux measurement.

Shallow groundwater could hypothetically be affected by either direct CO_2 interaction or brine displacement from a deeper reservoir. If brine were displaced, increases in calcium, magnesium, sodium, potassium, chloride, bromide, total dissolved solids, and specific conductivity would be expected. If CO_2 interacted directly with shallow groundwater, increases in alkalinity, dissolved CO_2, and decreases in pH would be expected. Changes in water chemistry would likely be first observed closest to the brine or CO_2 source.

Regulatory Compliance Wells

Four regulatory compliance shallow groundwater wells were drilled and constructed at IBDP during April and May 2010 as part of the 17-well network. The wells are about 140 feet (43 meters) deep with 10-foot (3 meter) screened intervals installed in thin sandstone of Pennsylvanian-age bedrock, the Illinois EPA-determined lowermost Underground Source of Drinking Water (USDW). Eleven key groundwater indicators are routinely monitored to determine if interactions are occuring between groundwater and brine or injected CO_2.

Monitored parameters include: pH, specific conductance, dissolved oxygen, temperature, groundwater elevation, alkalinity, bromide, calcium, chloride, sodium, and total dissolved CO_2. Analyses of the groundwater data to-date indicate there have been no observed changes in groundwater chemistry as a result of CO_2 injection. The natural variations observed in groundwater chemistry continue to help refine our conceptual understanding of local and regional shallow groundwater conditions. Research monitoring will conclude in 2017 after the three-year injection and three-year post-injection periods.



Conceptual model showing the shallow hydrogeological system at the IBDP site.



The soil gas sampling apparatus.

Initial development of a groundwater monitoring well.

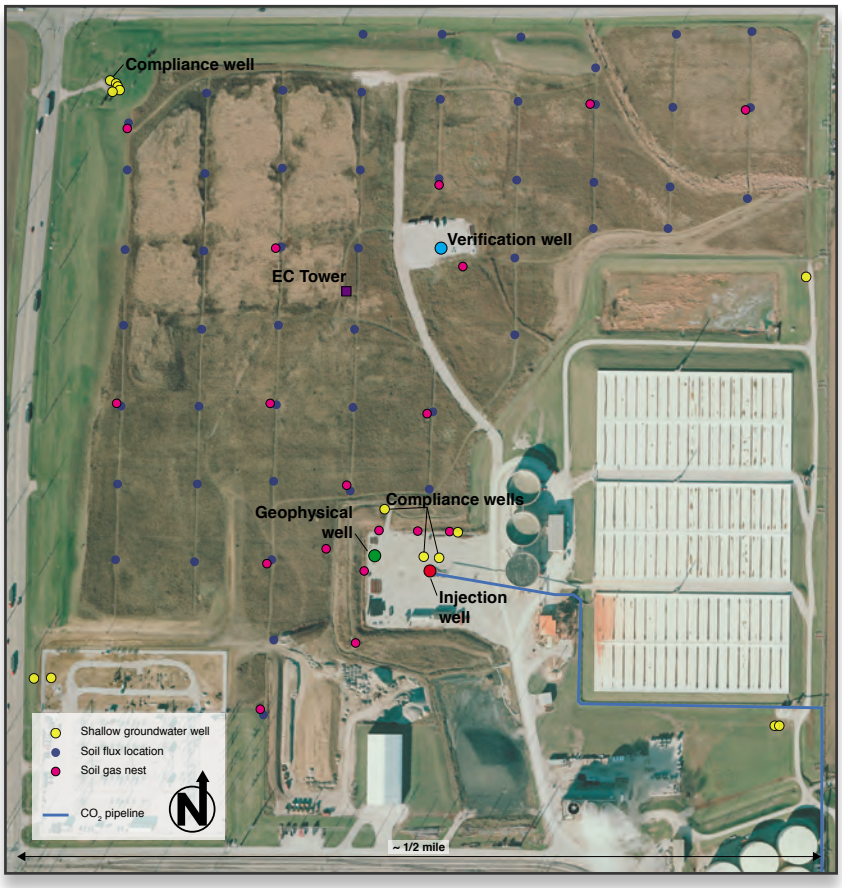
Collecting water samples from a regulatory compliance well.



The Illinois Basin - Decatur Project

The Midwest Geological Sequestration Consortium is conducting a large-scale CCS project in Decatur, Illinois, USA to demonstrate the ability of a deep saline formation to store one million tonnes of CO_2 from an ethanol production facility. In November 2011, CO_2 injection began at a rate of 1,000 tonnes/day and will continue for three years into the Mount Simon Sandstone at a depth of approximately 7,000 feet (2,100 meters). Injection is scheduled for completion in November 2014, at which time the project will move into the post-injection monitoring phase.





Illinois Basin - Decatur Project site and selected monitoring locations.

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